

Cisco Packet Tracer – Commands

show running-config

- Displays the current active configuration in RAM, including all settings and configurations that are currently applied to the device.

show startup-config

- Displays the configuration file stored in NVRAM, which is used to initialize the device when it boots.

configure terminal (conf t)

- Enters global configuration mode, allowing you to make changes to the device's configuration.

hostname [device-name]

- Sets the hostname of the device (e.g., hostname R1_123).

no ip domain-lookup

- Disables DNS lookup to prevent the device from attempting to resolve incorrectly entered commands.

enable secret [password]

- Sets a password for privileged EXEC mode access, which is encrypted for security (e.g., enable secret class).

line console 0

- Enters console line configuration mode for the physical console connection.

password [password]

- Sets a password for console access (e.g., password cisco).

login

- Enables password checking at login for the specified line (used in console and VTY lines).

line vty [0 4]

- Enters line configuration mode for virtual terminal lines (e.g., VTY 0-4 for SSH or Telnet access).

banner motd # [message] #

- Sets a Message of the Day (MOTD) banner to display to users when they log in (e.g., banner motd #Unauthorized access prohibited#).

interface [type] [number]

- Enters interface configuration mode for a specific interface (e.g., interface gigabitEthernet 0/1).

ip address [ip] [subnet mask]

- Assigns an IP address and subnet mask to an interface (e.g., ip address 194.164.40.161 255.255.255.0).

no shutdown

- Enables the interface (activates it).

crypto key generate rsa

- Generates an RSA key for SSH access. When prompted, enter the number of bits for the key (e.g., 2024).

ip domain-name [domain-name]

- Sets the domain name for the device (e.g., ip domain-name ccna-lab.com).

username [name] password [password]

- Creates a local user with a password (e.g., username SSHadmin password 55HAdm!n2020).

service password-encryption

- Encrypts all plaintext passwords in the configuration file.

ip ssh version 2

- Enables SSH version 2 on the device for secure remote access.

ip ssh time-out [seconds]

- Sets the SSH session timeout duration (e.g., ip ssh time-out 60).

ip ssh authentication-retries [number]

- Configures the number of authentication retries for SSH (e.g., ip ssh authentication-retries 3).

exec-timeout [minutes] [seconds]

- Configures the inactivity timeout for console or VTY lines (e.g., exec-timeout 5 0).

login block-for [seconds] attempts [number] within [seconds]

- Configures the router to block login attempts after a certain number of failed attempts (e.g., login block-for 120 attempts 3 within 60).

show ip interface brief

- Displays the status of all interfaces, including IP addresses and whether they are up or down.

interface range [type] [start-number] - [end-number]

- Configures a range of interfaces simultaneously (e.g., interface range gigabitEthernet 0/1 - 0/24).

shutdown

- Disables the interface (administratively down).

ping [ip address]

- Tests connectivity to a specific IP address by sending ICMP echo requests.

traceroute [ip address]

- Traces the route packets take to reach a specific IP address, displaying the hops along the path.

copy running-config startup-config

- Saves the current running configuration to the startup configuration stored in NVRAM.

show version

- Displays system information, including IOS version, memory, and hardware details.

show vlan brief

- Displays the VLAN configuration and associated ports.

show login

- Displays login status, showing if the device is in quiet mode due to failed login attempts.

reload

- Restarts the device, reloading the IOS and configuration.

enable

- Enters privileged EXEC mode.

show ip route

- Displays the routing table of the device, showing routes known to the router.

router ospf [process-id]

- Enters OSPF configuration mode for a specific process ID (e.g., router ospf 1).

network [network-address] [wildcard-mask] area [area-id]

- Configures a network for OSPF, assigning it to an area (e.g., network 192.168.1.0 0.0.0.255 area 0).

router bgp [as-number]

- Enters BGP configuration mode for a specific autonomous system (e.g., router bgp 65000).

neighbor [ip address] remote-as [as-number]

- Configures a BGP neighbor and specifies its autonomous system (e.g., neighbor 192.168.2.2 remote-as 65001).

default-information originate

- Configures the router to advertise a default route in OSPF.

show ip bgp

- Displays the BGP routing table, including all BGP routes and their status.

show ip ospf

- Displays OSPF status and configuration details.

show ip protocols

- Displays the status of routing protocols configured on the device, such as OSPF or EIGRP.